

Kapil Sharma

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SUMMARY

Software Engineer with **4 years and 8 months** of experience building scalable microservices and event-driven systems in production environments. Core expertise in Python, FastAPI, Django, AWS event-driven architecture, PostgreSQL/NoSQL data modeling, and production-grade **GenAI/LLM integrations** for workflow automation and decision support.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

Backend & Architecture: FastAPI, Django Rest Framework, Celery, Microservices, Event-Driven Architecture, REST APIs, WebSocket

Cloud & DevOps: AWS Lambda, ECS Fargate, SQS, EventBridge, CloudWatch, EC2, SES, Docker, Terraform, GitHub Actions

Databases & Caching: PostgreSQL, DynamoDB, MongoDB, Redis, MySQL, pgvector

AI & LLM Engineering: LangChain, LangGraph, OpenAI API, RAG Pipelines, Vector Embeddings, LiteLLM, RAGAS, MCP

Protocols & Messaging: REST, WebSocket, MQTT, WebRTC, RabbitMQ, SQS

WORK EXPERIENCE

Tiger Analytics

Senior Application Engineer

Oct 2024 – Present
Bangalore, India (Hybrid)

- Designed and owned an event-driven automated email system using **AWS EventBridge** and Amazon SES, triggered every 15 minutes to process incremental claim updates and reliably deliver approx. 100K emails per day.
- Designed, developed, and maintained **FastAPI-based microservices** supporting high-throughput claim processing in a production auto-adjudication environment.
- Built backend workflows using **AWS Lambda, SQS, ECS Fargate**, CloudWatch, and EC2 to improve scalability, observability, and operational stability.
- Optimized PostgreSQL schemas and DynamoDB data models for rule metadata, claim evidence, and audit logs while applying TDD with 90%+ unit and integration test coverage.
- Integrated **LLM-based decision-support components** using structured prompts and retrieval-augmented context to assist complex claim validation and decisioning workflows.
- Reviewed code contributions, contributed to **CI/CD pipelines** and **infrastructure-as-code** workflows, and collaborated with cross-functional teams on API contracts, test plans, and delivery milestones.

VVDN Technologies

Software Engineer

Sept 2021 – Oct 2024
Gurugram, India

- Designed and maintained **REST APIs** using **Django Rest Framework**, improving throughput by 15–17% through SQL query optimization and backend refactoring.
- Developed backend features for Netgear's cloud-native Insight platform using Python, PostgreSQL, and MongoDB to manage and orchestrate thousands of edge devices.
- Built real-time IoT integrations for NXP Home Assistant using MQTT and implemented **WebRTC signaling** for low-latency video streaming dashboards.
- Collaborated with global engineering teams to debug and resolve production issues, improving system reliability and service availability.
- Built an internal HR chatbot using **OpenAI APIs** and **RAG** over company policy documentation, while mentoring junior engineers and contributing to code reviews.

PERSONAL PROJECTS

ShipGuard - AI Release Risk Reviewer

Python, Typer, GitHub API, OpenAI-compatible APIs, Pydantic, PyPI

[GitHub](#) | [PyPI](#)

- Built and published a **PyPI-installable developer tool**, **shipguard-ai**, that analyzes local git diffs and GitHub pull requests to detect release risks beyond passing CI, including API, database, config, dependency, rollback, and test-coverage risks.
- Implemented PR ingestion, risk-aware diff prioritization, repository memory, **structured LLM-based risk reports**, Markdown/JSON/HTML release artifacts, and optional PR summary/inline review comments.

SMARN - Self Memory Archive & Recall Network

Python, FastAPI, PostgreSQL, pgvector, SQLAlchemy, Telegram Bot, OpenAI

[GitHub](#)

- Built a **Python-based personal memory system** that stores Telegram text and voice memories in **PostgreSQL with pgvector**, enriches memories using LLMs, retrieves relevant context, and answers user questions through natural recall.
- Designed backend components with SQLAlchemy, Alembic, Pydantic, Docker Compose, embeddings, semantic search, daily/weekly reviews, voice-note processing, and Telegram commands such as **/remember**, **/ask**, and **/weekly_review**.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

B.Tech in Computer Science & Engineering

2017 – 2021

GPA: 8.58